WORKED SOLUTION

STEP III 1995, Question 5

Pure • Maclaurin Series

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s3-q5

The key idea

Write $\theta = \sin^{-1} x$, so that $\mathrm{d}\theta/\mathrm{d}x = 1/\sqrt{1-x^2}$. Multiplying $y^{(1)}$ by $\sqrt{1-x^2}$ cancels that awkward factor, and differentiating once more gives the second-order equation with no square roots in it. Leibniz's rule then turns the equation into a recurrence, and at $x = 0$ the factor $1 - x^2$ becomes 1 and the term in x disappears. The recurrence then hands us every derivative at 0.

Solution

The differential equation

Let $\theta = \sin^{-1} x$, so $\sin\theta = x$ and, for $|x| < 1$, $\cos\theta = \sqrt{1-x^2} > 0$, with $\dfrac{\mathrm{d}\theta}{\mathrm{d}x} = \dfrac{1}{\sqrt{1-x^2}}$.

Then $y = \sin^2 m\theta$, and by the chain rule

$$y^{(1)} = 2m \sin m\theta \cos m\theta \cdot \frac{1}{\sqrt{1-x^2}} = \frac{m \sin 2m\theta}{\sqrt{1-x^2}}, \qquad \text{so} \qquad \sqrt{1-x^2}\, y^{(1)} = m \sin 2m\theta.$$

Differentiate this with respect to x, using the product rule on the left:

$$\sqrt{1-x^2}\, y^{(2)} - \frac{x}{\sqrt{1-x^2}}\, y^{(1)} = 2m^2 \cos 2m\theta \cdot \frac{1}{\sqrt{1-x^2}}.$$

Multiply through by $\sqrt{1-x^2}$, and use $\cos 2m\theta = 1 - 2\sin^2 m\theta = 1 - 2y$:

$$(1-x^2)\, y^{(2)} = x\, y^{(1)} + 2m^2(1-2y),$$

which is the required equation.

Ask yourself Why not square $\sqrt{1-x^2}\, y^{(1)} = m \sin 2m\theta$ first?

Squaring gives $(1-x^2)(y^{(1)})^2 = 4m^2 y(1-y)$, and differentiating that leaves a factor $2y^{(1)}$ on every term. We would then have to divide by $y^{(1)}$, which is zero at $x = 0$,

the very point we care about later. Differentiating before squaring avoids the division altogether.

The recurrence

Differentiate the equation n times, with $n \geq 1$, using Leibniz's rule. Since $1 - x^2$ has only two non-zero derivatives ($-2x$ and -2), and x has only one,

$$\begin{aligned}
\frac{\mathrm{d}^n}{\mathrm{d}x^n}\left[(1-x^2)y^{(2)}\right] &= (1-x^2)y^{(n+2)} - 2nx\,y^{(n+1)} - n(n-1)\,y^{(n)},\\
\frac{\mathrm{d}^n}{\mathrm{d}x^n}\left[x\,y^{(1)}\right] &= x\,y^{(n+1)} + n\,y^{(n)},\\
\frac{\mathrm{d}^n}{\mathrm{d}x^n}\left[2m^2(1-2y)\right] &= -4m^2\,y^{(n)}.
\end{aligned}$$

The last line needs $n \geq 1$, because only then does the constant $2m^2$ differentiate to zero. Equating the two sides,

$$(1-x^2)y^{(n+2)} - 2nx\,y^{(n+1)} - n(n-1)y^{(n)} = x\,y^{(n+1)} + n\,y^{(n)} - 4m^2y^{(n)},$$

and collecting terms ($2n+1$ lots of $xy^{(n+1)}$, and $n(n-1)+n = n^2$) gives

$$(1-x^2)\,y^{(n+2)} = (2n+1)\,x\,y^{(n+1)} + (n^2 - 4m^2)\,y^{(n)} \qquad (n \geq 1),$$

as required.

The Maclaurin series

At $x = 0$ we have $\theta = 0$, so

$$y(0) = \sin^2 0 = 0, \qquad y^{(1)}(0) = m\sin 0 = 0, \qquad y^{(2)}(0) = 0 + 2m^2(1-0) = 2m^2,$$

the last from the differential equation. Putting $x = 0$ in the recurrence gives

$$y^{(n+2)}(0) = (n^2 - 4m^2)\,y^{(n)}(0) \qquad (n \geq 1).$$

Odd derivatives. Since $y^{(1)}(0) = 0$, the recurrence gives $y^{(3)}(0) = 0$, then $y^{(5)}(0) = 0$, and so on: every odd derivative vanishes at 0. That makes sense, because y is an even function of x.

Even derivatives. Take $n = 2j$:

$$y^{(2j+2)}(0) = (4j^2 - 4m^2)\,y^{(2j)}(0) = -4(m^2 - j^2)\,y^{(2j)}(0) \qquad (j \geq 1).$$

Starting from $y^{(2)}(0) = 2m^2$ and applying this for $j = 1, 2, \ldots, k-1$,

$$y^{(2k)}(0) = 2m^2 \prod_{j=1}^{k-1}(-4(m^2 - j^2)) = (-1)^{k-1}\,2^{2k-1}\prod_{j=0}^{k-1}(m^2 - j^2),$$

where we absorbed the leading m^2 as the $j = 0$ factor. (For $k = 1$ the formula gives $2m^2$, as it should.)

Hence the Maclaurin series is

$$y = \sin^2(m \sin^{-1} x) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}\, 2^{2k-1}}{(2k)!}\, m^2(m^2-1^2)(m^2-2^2)\cdots(m^2-(k-1)^2)\, x^{2k},$$

and its first few terms are

$$y = m^2x^2 - \frac{m^2(m^2-1)}{3}\, x^4 + \frac{2m^2(m^2-1)(m^2-4)}{45}\, x^6 - \cdots.$$

If m is an integer, every term with $k > |m|$ contains the factor $m^2 - m^2 = 0$, so the series stops and y is a polynomial of degree $2|m|$.

Checks

- $m = 1$: $y = \sin^2(\sin^{-1} x) = x^2$, and the series gives x^2 and then zeros. $m = 2$: the series gives $4x^2 - 4x^4$, and indeed $\sin^2 2\theta = 4\sin^2\theta(1 - \sin^2\theta) = 4x^2 - 4x^4$.
- $m = \frac{1}{2}$, $x = \frac{1}{2}$: $y = \sin^2(\pi/12) = \frac{1}{2}\left(1 - \frac{\sqrt{3}}{2}\right) \approx 0.066987$. The series is $\frac{1}{4}x^2 + \frac{1}{16}x^4 + \frac{1}{32}x^6 + \cdots$, whose first three terms give 0.066895, and adding more terms closes the gap.
- We checked numerically that the differential equation holds at random x and m, that the series coefficients satisfy it exactly, and that the series (to 300 terms) agrees with y to about 10^{-15} for $|x| \le 0.8$.

Common slips

- Using the recurrence with $n = 0$. It holds only for $n \ge 1$, because the constant $2m^2$ survives when $n = 0$. The case $n = 0$ is the original equation, which is how we got $y^{(2)}(0) = 2m^2$.
- Getting the Leibniz coefficients wrong: the term $-n(n-1)y^{(n)}$ comes from $\binom{n}{2} \cdot (-2)$, not from $\binom{n}{2} \cdot (-2x)$.
- Writing the general term without the alternating sign. Each step multiplies by $4(j^2 - m^2)$, which is negative when $m^2 > j^2$.

Beyond STEP

A Maclaurin series built from derivatives might not equal the function it came from, but here it does for $|x| < 1$. The series has radius of convergence 1 (by the ratio test, when m is not an integer), and $\sin^{-1} x$ stops being well behaved at $x = \pm 1$. The polynomials you get for integer m are relatives of the Chebyshev polynomials, since $\cos 2m\theta$ is a polynomial in $\sin\theta$. To see why such series really do converge to their functions, read about Taylor's theorem with remainder in Abbott, *Understanding Analysis*. Needham's *Visual Complex Analysis* explains why the radius of convergence is set by the trouble at ± 1.

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